OCTO-EARTH

Reinforcement Learning Technical Report

Evidence-based review of the PPO controller and its planetary-propulsion simulation

EXECUTIVE FINDING

The saved best PPO checkpoint completes the configured high-fidelity mission and passes a 200-year ballistic stability check. However, the 1,000-episode stochastic training log contains zero successful episodes, and every periodic deterministic evaluation after episode 50 fails. The hand-designed heuristic is faster, uses fewer resources, and requires fewer Mars-shield overrides. The result demonstrates a functioning simulated control loop, not stable policy learning or RL superiority.

At a glance

Metric	Best PPO checkpoint	Interpretation
High-fidelity outcome	Success; validation passed	One high-fidelity case, seed 2001
Final orbit	2.6972 AU; e = 0.00748	Target 2.70 AU +/- 0.05 AU; e <= 0.05
Time to target	7,200 simulated years	Within the 10,000-year deadline
Surface climate	286.0 K final; 4.68 K max deviation	Passes +/- 5 K; only 0.32 K max-deviation margin
Mars safety	0.04928 AU closest pass; 472 overrides	Above 0.02456 AU configured safe distance
Training outcome	0 / 1,000 stochastic episodes successful	Selected checkpoint came from a separate deterministic evaluation

Mission and claim boundary

The environment models moving Earth from 1 AU to 2.7 AU within 10,000 simulated years while respecting orbital, Moon-retention, Mars-safety, climate and resource constraints. The configuration is explicitly marked “hypothetical”. All results are conditional on idealized enablers, including an enormous asteroid mass driver, a solar collector and microwave power network, and space mirrors. This is a simulation result, not a demonstration of buildable hardware.

Evidence cut-off: 25 October 2026. Primary run evidence: comparison.json, evals.json, training_log.csv, best_model_info.json, physics_checks.json, config.json and feasibility.json. The uploaded report.md is stale and is not treated as the authoritative run record.

1. Problem definition and end-to-end architecture

The controller chooses how to eject asteroid material so recoil changes the Earth-Moon barycentre orbit. The agent does not directly overwrite Earth velocity. The action decoder turns normalized commands into packet mass, launch speed, thrust direction in the rotating radial-transverse-normal (RTN) frame, and a climate-light set-point. The physics engine applies the reaction, charges energy, accounts for packet fate, advances Mars interactions and returns a new observation and reward.

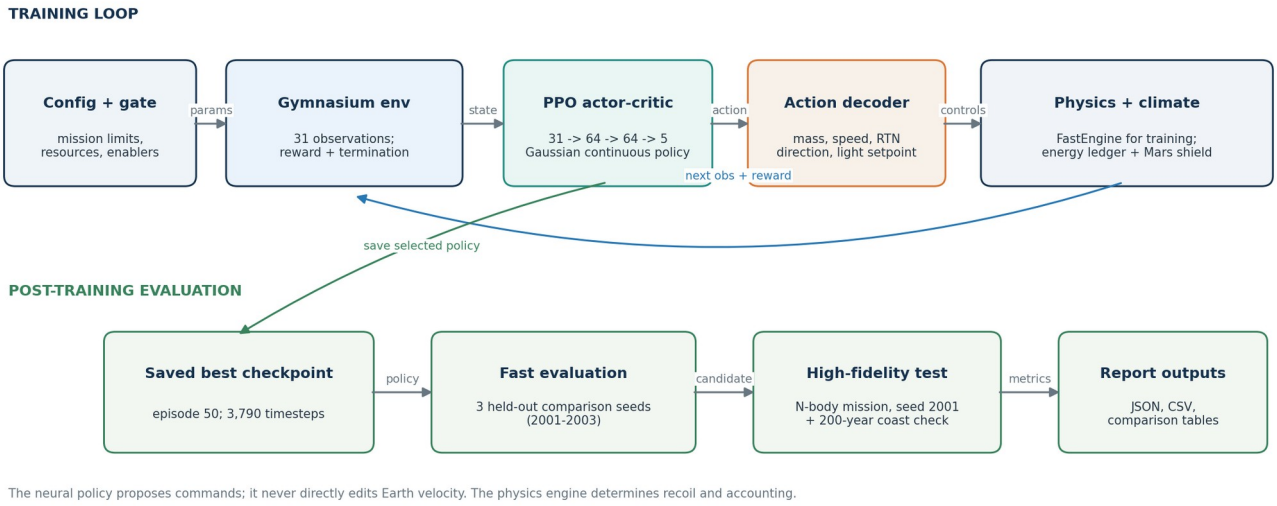


Figure 1. Training loop and separate post-training validation path. The high-fidelity model evaluates a saved policy; it does not train PPO.

1.1 Main components

Component	Role in the implementation	Evidence boundary
Config + feasibility gate	Mission limits, resources, target tolerances, collector/mirror assumptions; analytic checks before training.	Analytic feasibility is conditional on the preset.
OctoEarthEnv (Gymnasium)	31-value observation, 5-value continuous action, reward decomposition and termination.	Defines the control task; does not certify real engineering.
FastEngine	Fast physics used for PPO rollouts; interval-level orbital and resource updates.	Training fidelity is lower than high-fidelity validation.
Safety shield	Forecasts Mars close approaches and can override candidate commands.	Successful missions include shield intervention; report overrides.
HifiEngine	N-body validation with Sun, Earth, Moon, Mars and fragment accounting.	One comparison seed per policy in this run.
Climate + energy ledger	Surface/deep-ocean thermal response, mirror light, collector power and energy bookkeeping.	Depends on declared hypothetical enablers and parameters.
Evaluation/report pipeline	Saves checkpoint metadata, periodic evaluations, CSV logs and policy comparison JSON.	Raw artifacts expose training collapse missed by stale report.md.

1.2 Mission constraints used in this run

Parameter	Configured value
Target orbit	2.70 AU; semi-major axis tolerance +/- 0.05 AU
Orbital shape / tilt	eccentricity <= 0.05; inclination <= 5 degrees
Deadline / decision interval	10,000 years maximum; one decision per 100 years (up to 100 decisions)
Propulsion reservoir	3.0 x 10^22 kg; max packet 1.0 x 10^18 kg; 8 launches/year
Launch speed / energy reference	20,000 to 5,000,000 m/s; configured energy budget 3.0 x 10^35 J
Climate target	288 K +/- 5 K; livable band monitored as +/- 15 K
Mars safety	2,000,000 km hard minimum plus tide and Moon-orbit-perturbation constraints

2. Markov decision process

2.1 Episode and observation

One episode is a full mission. Every action is held over a 100-year simulated interval, with up to 800 packet-launch events per step at the configured rate. The observation is a clipped/scaled float32 vector of 31 continuous values. The current source and both saved model archives agree on observation shape (31,) and action shape (5,).

Observation group	Indices	Count	Information supplied
Earth-Moon barycentre state	0-5	6	Position (3) and velocity (3), scaled.
Orbit and mission/resource state	6-16	11	Semi-major axis, eccentricity, inclination, energy gap, Moon distance/bound state, remaining mass/energy/time, estimated mass and energy capability.
Previous action	17-21	5	Previous normalized controls, allowing the policy to observe its last command.
Mars safety context	22-26	5	Relative Mars position, current distance relative to safe distance, and predicted coast closest approach over the shield horizon.
Climate and mirrors	27-30	4	Surface and deep-ocean temperature error, sunlight share, available mirror capacity.

2.2 Continuous action space

Action	Meaning	Decoding
a0	Ejection throttle	Maps from [-1,1] to 0-100% of max packet mass; values below the 2% throttle threshold produce no ejection.
a1	In-plane angle	Angle in the RTN plane, from prograde toward radial.
a2	Out-of-plane angle	Cross-plane thrust angle.
a3	Launch speed	Linear map between 20 km/s and 5,000 km/s.
a4	Climate set-point	Requests 0.5 to 1.5 times today's sunlight; mirrors add light up to capacity but do not subtract natural sunlight.

2.3 Reward, not just the final score

The reward is the sum of terms logged by the environment. The progress term uses change in an orbital-energy potential. Other terms penalize eccentricity/inclination outside limits, energy and mass use, time, Mars-safety proximity and climate deviation. A terminal success bonus is added after the success conditions pass; terminal orbit error and hard-failure penalties apply at termination.

Reward component	Default weight	Design intent
Progress; success	100; 100	Reward movement toward target orbital energy and task completion.
Energy use; mass use	10; 5	Discourage resource consumption.
Eccentricity; inclination	1; 1	Penalize orbit shape/tilt outside allowed thresholds.
Time; Mars safety; climate	0.02; 20; 2	Add time cost and discourage unsafe or climate-degrading trajectories.
Terminal error; failure	50; 100	Penalize missing the target or hard safety/mission failure.

Important interpretation

The run exposes a reward trade-off: later evaluations conserve resources by applying almost no useful propulsion, miss the deadline, and still receive less-negative rewards. This suggests reward-objective misalignment or policy collapse; the logs do not isolate one specific implementation bug.

3. PPO setup and training results

3.1 Algorithm and saved settings

The policy uses Proximal Policy Optimization (PPO) in Stable-Baselines3. A two-hidden-layer multilayer perceptron with 64 units per layer is used for both actor and critic. PPO models a Gaussian continuous-action policy; its critic estimates expected return. Training uses the fast environment and one vectorized environment instance.

Setting	Saved value
Algorithm / network	PPO; MLP [64, 64]; tanh activations in the SB3 policy
n_steps / batch size / epochs	1,000 / 200 / 10
Learning rate	3 x 10^-4
Discount / GAE lambda	gamma = 0.995; lambda = 0.95
Initial action log standard deviation	-0.5
Entropy coefficient / seed	0.0 / 1
Configured episode target	1,000 episodes; evaluation every 50 episodes

3.2 What the output files actually show

Artifact	Observed fact	Meaning
training_log.csv	1,000 stochastic episodes; 94,586 timesteps; 0 successful episodes.	Exploration episodes never completed the mission.
Termination reasons	641 deadline; 359 resources_exhausted.	Every training episode failed.
evals.json	20 periodic deterministic evaluations. Episode 50 succeeds; episodes 100-1,000 all show 0 success.	The best result is early and does not persist.
best_model.zip + metadata	Selected at episode 50, 3,790 timesteps; deterministic eval success 100% on seeds 1001-1002.	Different from stochastic episode 50, which failed.
last_model.zip	94,586 timesteps; episode-1,000 evaluation has a=1.00198 AU, 10,000 years, energy fraction 0.0017 and success 0.	Later policy is poor; do not use as demo model.

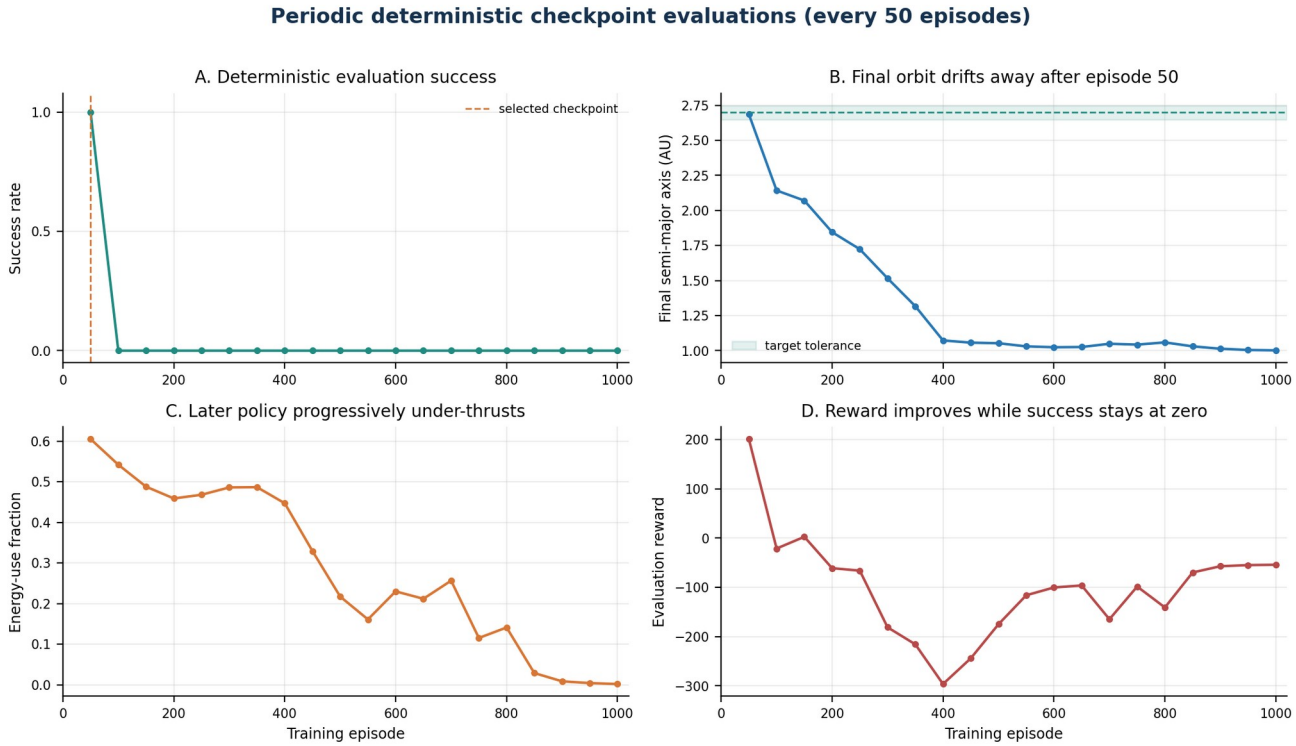


Figure 2. Periodic deterministic evaluations. After the early selected checkpoint, success drops to zero while the policy approaches a near-no-op strategy.

The training episode at episode 50 itself has success=0 and reason=resources_exhausted. The saved best model's separate deterministic evaluation at the same training milestone succeeds; these are different experiments and must not be conflated.

4. Policy comparison: high-fidelity seed 2001

The comparison procedure runs the fast model on seeds 2001-2003, then one high-fidelity N-body mission on seed 2001 followed by a 200-year ballistic validation. Mission success requires the high-fidelity mission and stability check to pass. The fast model's three-seed success rate is useful but is not a substitute for multi-seed high-fidelity validation.

Policy	Fast success	Hi-fi outcome	Years	Final a (AU)	e	Mass used	Energy used
No ejection	0/3 (0%)	Deadline; validation fails	10,000	1.0000	0.01593	0%	0 J
Hand-designed heuristic	3/3 (100%)	Success; validation passes	3,600	2.7000	0.00889	95.25%	1.757 x 10^35 J
Best PPO	3/3 (100%)	Success; validation passes	7,200	2.6972	0.00748	97.28%	1.839 x 10^35 J

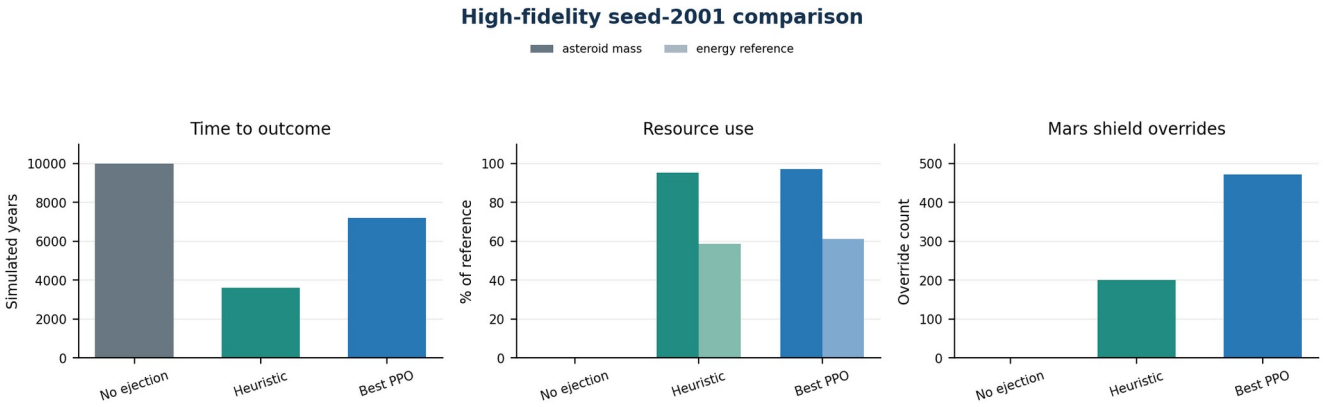


Figure 3. The heuristic is twice as fast as PPO and uses fewer resources. The override count includes the active Mars shield.

The energy bars are normalized to the configured 3.0 x 10^35 J budget reference; collector-source energy and reserve draw are detailed on page 6.

4.1 Safety and climate comparison

Metric (high-fidelity)	No ejection	Heuristic	Best PPO
Closest Mars pass	0.37211 AU	0.05005 AU	0.04928 AU
Configured hard safe distance	0.02456 AU	0.02456 AU	0.02456 AU
Mars-shield overrides	0	200	472
Final surface temperature	288.17 K	288.33 K	286.00 K
Maximum climate deviation	2.08 K	2.57 K	4.68 K
Years outside +/-15 K livable band	0	0	0
Moon bound at mission end	Yes	Yes	Yes
200-year orbit validation	Fail: target not reached	Pass	Pass

Primary comparison conclusion

PPO reaches the target under this simulation, but it takes 3,600 additional years, uses about 4.7% more propulsion energy and 2.0 percentage points more asteroid mass than the heuristic, and triggers 472 vs 200 Mars-shield overrides. The heuristic is the stronger current baseline.

5. High-fidelity mission behaviour

High-fidelity trajectories; climate remains dependent on assumed mirrors

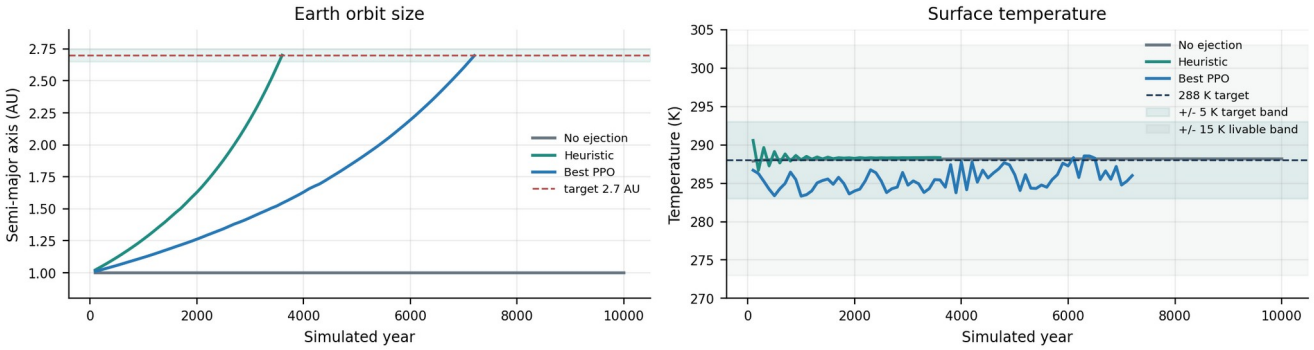


Figure 4. Semi-major axis and surface-temperature histories from the saved high-fidelity trajectories. The climate trajectory depends on the modeled space-mirror system.

5.1 Orbit, Moon, Mars and safety shield

- The best PPO run reaches a=2.69719 AU, eccentricity 0.00748 and inclination 0.2616 degrees after 7,200 years. The 200-year coast check reports max |delta-a| = 0.00282 AU, max eccentricity 0.00748 and a bound Moon; all validation conditions pass.
- The closest Earth-Mars distance is 0.04928 AU (about 7.37 million km), exceeding the configured 0.02456 AU safety distance. The heuristic's closest pass is 0.05005 AU. Both missions pass the recorded Mars safety checks.
- PPO's 472 shield overrides are 2.36 times the heuristic's 200. Therefore some of the successful safety outcome is supplied by the shield rather than demonstrated as an independently learned safe policy. Mars's long-term orbit is chaotic in the project model.

5.2 Climate and energy systems

Output	Heuristic	Best PPO
Final surface / deep-ocean T	288.33 / 288.34 K	286.00 / 285.74 K
Max climate deviation	2.57 K	4.68 K
Mirror light onto Earth	1.040 x 10^28 J	1.947 x 10^28 J
Surface T without mirrors (model estimate)	about 157 K	about 155 K
Collector-sourced mission energy	1.757 x 10^35 J	1.839 x 10^35 J
Reserve energy used	0 J	0 J

PPO's maximum climate deviation is only 0.319 K inside the configured +/-5 K target tolerance, versus 2.434 K of margin for the heuristic. Both show zero years outside the wider +/-15 K livable band. Without the assumed space mirrors, the model predicts severe cooling at 2.7 AU; habitability is not guaranteed by orbital transfer alone.

5.3 Collector belt maintenance

With station keeping enabled, the PPO collector-belt summary reports a maximum deviation of about 386 km (RMS 317 km) and roughly 51 m/s/year control delta-v. Without station keeping, its reported drift reaches about 130 million km, or roughly 2,009 collector slots. This is a large ongoing control requirement, not a passive stable ring.

6. Feasibility gate, physics checks and declared enablers

6.1 Analytic feasibility is a model gate, not a technology claim

Analytic quantity	Saved value	Interpretation
Slow-spiral delta-v, 1 to 2.7 AU	11,658 m/s	Hohmann reference is 10,997 m/s.
Earth-Moon orbital energy change	1.697 x 10^33 J	Approximately 5.38 x 10^21 W averaged over 10,000 years.
Minimum launch energy estimate	1.664 x 10^35 J	Within the modeled reserve + collector supply.
Available energy over mission	3.817 x 10^36 J	Includes 3 x 10^35 J reserve plus assumed solar collector power.
Reaction mass / reservoir	1.415 x 10^22 kg at max speed; 3 x 10^22 kg available	Combined mass/time limit uses all 3 x 10^22 kg at 2.355 x 10^6 m/s.
Minimum transfer time at limit	1,771 years	Below the 10,000-year deadline.
Collector output at 2.7 AU	1.114 x 10^25 W electrical	Model states 1.76x the maximum launcher peak demand.

6.2 Recorded automated physics self-checks

The uploaded physics_checks.json contains 16 recorded checks, all marked passed. Coverage includes: two-body/Kepler agreement; gravity contributions; equal-and-opposite forces; momentum and angular-momentum accounting; propulsion energy accounting; finite mass and fragment accounting; prograde/retrograde sign; fallback/escape/impact outcomes; fixed-seed reproducibility; viewer-state continuity; Moon behaviour in the fast model; two-way Mars gravity; Mars safety limits and shield; solar collection/microwave chain; climate, mirrors and energy ledger; and collector-belt station keeping.

Numerical residuals are not interchangeable

In the saved PPO high-fidelity mission, the energy-ledger residual is about 4.07 x 10^-16, while the separate whole-system energy-conservation residual is about 1.52 x 10^-6. Momentum and angular-momentum residuals are about 4.0 x 10^-13 and 1.9 x 10^-13. Report each diagnostic by its name; do not state that all conservation errors are at machine precision.

6.3 Hypothetical enablers that dominate feasibility

- E1 - planetary launcher: assumes packets up to 10^18 kg, launch speeds up to 5,000 km/s, eight launches per year, 50% kinetic efficiency and heat rejected away from Earth. The hardware, launch shock, Earth damage and atmospheric effects are not fully modeled.
- E2 - solar collector and microwave network: assumes interception of 30% of solar output at 0.5 AU, millions of large collectors and high efficiency transfer/capture. Construction, station keeping and deployment are idealized at this scale.
- E3 - space mirrors: assumes about 1.6 x 10^15 m^2 of mirrors, 90% reflectivity and the geometry needed to supply extra sunlight at 2.7 AU. The climate result relies on this assumption.

The 16 checks are project self-checks visible in the saved artifact. They should not be claimed to equal the participant guide's complete official benchmark suite without a separate benchmark-by-benchmark mapping and run record.

The saved physics_checks.json does not include recorded results for the source-defined optional time-step convergence and 100-year no-thrust energy-drift checks. Do not claim those checks passed from this artifact alone.

7. Discussion, limitations and recommended next experiments

7.1 Conclusions supported by the run

- A saved PPO checkpoint can complete the configured fast evaluation and one high-fidelity mission, including a 200-year orbit-stability check, while the Moon remains bound and the model stays within configured Mars and climate limits.
- The PPO training process did not produce reliable learning in this run: all 1,000 stochastic episodes failed; only the separate deterministic checkpoint evaluation at episode 50 succeeded; all 19 later periodic evaluations failed.
- The heuristic is the stronger current policy: it completes the mission in half the time, uses less modeled propulsion energy and asteroid mass, gives more climate margin, and requires fewer safety-shield overrides.
- The simulation's feasibility result is conditional on extreme declared hypothetical enablers and should not be presented as real-world engineering feasibility.

7.2 Priority fixes / experiments

Priority	Next action	Evidence required before claiming improvement
1 - reward diagnosis	Log every reward component for successful best-checkpoint evals and late failing evals; review energy/mass penalties versus deadline and success bonuses.	Explain why avoiding resource use produces a less-negative return than useful progress; change weights only with a stated hypothesis.
2 - preserve baseline/checkpoints	Keep best_model.zip separate from last_model.zip and keep heuristic/no-ejection outputs unchanged.	Automated comparison on the same seeds and a machine-readable results table.
3 - generalization	Evaluate best PPO and heuristic on at least 20-30 unseen randomized phases; use multiple high-fidelity seeds where compute permits.	Success rate and spread for years, resources, climate margin, Mars distance and override count.
4 - stress-test shield/climate	Run ablations with shield off only in safe test scenarios; test lower climate margins and collector availability.	Separate learned-policy effects from guardrail effects and test sensitivity to hypothetical assumptions.
5 - submission validation	Map each required participant-guide benchmark to a test name and captured result; rerun on final code revision.	Versioned test report, physics settings, source commit/archive hash and exact result artifacts.

7.3 Reproducibility record and evidence index

Artifact	Use in this report
config.json	Canonical scenario parameters and reward weights.
best_model_info.json + best_model.zip	Checkpoint episode, timestep, hyperparameters and selection rule.
last_model.zip	Final training checkpoint; not recommended for current demonstration.
training_log.csv + evals.json	Stochastic episode outcomes and the 20 deterministic evaluations.
comparison.json	Fast 3-seed results, high-fidelity seed-2001 run, trajectory traces, climate, Mars, energy, belt and validation metrics.
physics_checks.json + feasibility.json	Saved self-check outcomes and analytic mass/energy/rate gate.
octoearth_rl.py + RL_CODE_GUIDE.md	Current environment, observation/action encoding, PPO setup and comparison procedure.
Participant guide PDF	Mission requirements, conservation/safety/habitability goals, benchmark expectations and report limit.

Submission wording recommended

"The selected PPO checkpoint completes the modeled high-fidelity mission under declared hypothetical enablers, but the recorded training run exhibits policy collapse after the early selected checkpoint. The hand-designed heuristic remains the stronger baseline on mission time, resource use and shield interventions."

Source note: Results are transcribed from the uploaded run artifacts created 25 October 2026; implementation claims are checked against the supplied octoearth_rl.py and RL_CODE_GUIDE.md. The supplied report.md contained stale training totals and a "physics checks not run" statement, so it was not used as the authoritative run summary.